

Practical no. 2

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Class :- CO3K – B

Subject :- CGR

Aim :- Write a C program to draw a line using DDA algorithm

Program :-

```
#include<stdio.h>

#include<conio.h>

#include<graphics.h>

void main()

{

int gd=DETECT,gm,i;

float x,y,dx,dy,steps;

int x0,x1,y0,y1;

initgraph(&gd,&gm,"C:\\\\TurboC3\\\\BGI");

setbkcolor(WHITE);

x0=100,y0=200,x1=500,y1=300;

dx=(float)(x1-x0); //slope of line

dy=(float)(y1-y0); //slope of line

if(dx>=dy)

{

steps=dx;

}

else

{

steps=dy;
```

```
}  
  
dx=dx/steps;  
  
dy=dy/steps;  
  
x=x0;  
  
y=y0;  
  
i=1;  
  
while(i<=steps)  
{  
  
putpixel(x,y,BLUE);  
  
x+=dx;  
  
y+=dy;  
  
i=i+1;  
  
}  
  
getch();  
  
closegraph();  
  
}
```

OUTPUT :-

